

# Project Design Phase

## Solution Architecture

**Date:** 18 June 2026

**Project Name:** Online Complaint Registration System

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## Solution Architecture Overview

The Online Complaint Registration System is designed using the MERN Stack architecture. The solution consists of three major layers: Frontend Layer, Backend Layer, and Database Layer.

The architecture enables secure communication between users, administrators, and the database through REST APIs, ensuring efficient complaint registration, tracking, management, and resolution.

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## Architecture Components

### Frontend Layer (React.js)

The frontend provides the user interface for:

- User Registration and Login
  - Complaint Registration
  - Complaint Tracking
  - Complaint History
  - Notifications
  - Dashboard Management
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### Backend Layer (Node.js & Express.js)

The backend handles:

- API Requests
- Authentication & Authorization

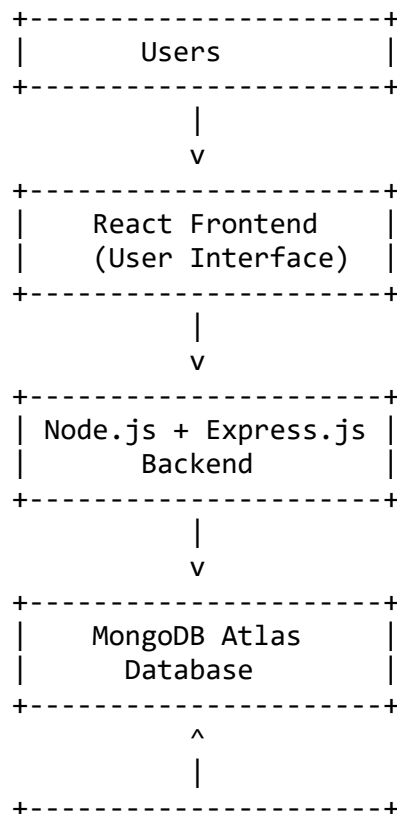
- Complaint Processing
  - Complaint Status Management
  - Notification Management
  - User Management
  - Administrative Operations
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## Database Layer (MongoDB Atlas)

The database stores:

- User Information
  - Complaint Records
  - Complaint Status Details
  - Notification Data
  - Administrative Records
  - System Logs
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## Solution Architecture Diagram



| Administrators |  
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## Data Flow

### Step 1

User logs into the application.

### Step 2

User submits a complaint through the frontend interface.

### Step 3

Frontend sends requests to backend APIs.

### Step 4

Backend validates requests and processes complaint data.

### Step 5

MongoDB stores and retrieves complaint information.

### Step 6

Administrators review and update complaint status.

### Step 7

The system sends status updates and notifications to users.

### Step 8

Users track complaint progress through the dashboard.

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## Benefits of the Architecture

- Scalable and modular design.
- Secure authentication using JWT.
- Efficient complaint management.
- Centralized complaint database.
- Real-time complaint tracking.

- Improved communication between users and administrators.
  - Easy deployment and maintenance.
  - Support for future enhancements.
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## Technologies Used

### Frontend

- React.js
- Vite
- Axios
- React Router DOM

### Backend

- Node.js
- Express.js
- JWT Authentication
- bcryptjs

### Database

- MongoDB Atlas
- Mongoose ODM

### Deployment

- Vercel
  - Render
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## Conclusion

The proposed MERN Stack architecture provides a secure, scalable, and efficient solution for complaint management. It ensures smooth communication between users, administrators, and the database while supporting complaint registration, tracking, monitoring, and resolution.

The architecture improves transparency, efficiency, accountability, and user satisfaction while supporting future scalability and advanced features.